

Write a C program to find cube of any number using function.

```
#include <stdio.h>

double cube(double num);

int main()
{
    int num;

    double c;

    printf("Enter any number: ");

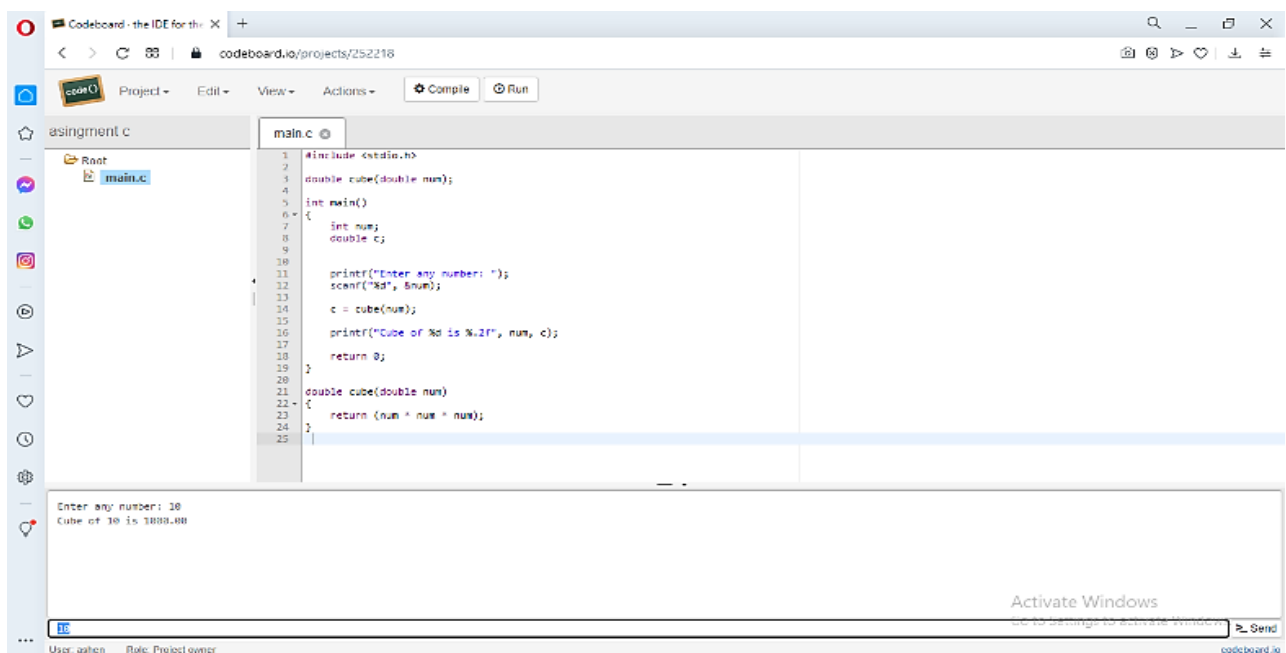
    scanf("%d", &num);

    c = cube(num);

    printf("Cube of %d is %.2f", num, c);

    return 0;
}

double cube(double num)
{
    return (num * num * num);
}
```



The screenshot shows the CodeBoard IDE interface. The main editor displays the C program code, which is identical to the one provided in the previous block. The left sidebar shows the project structure with a file named 'main.c'. The bottom console area shows the output of the program: 'Enter any number: 10' followed by 'Cube of 10 is 1000.00'. The status bar at the bottom indicates the user is 'anthon' and the role is 'Project owner'.

Write a C program to find diameter, circumference and area of circle using functions.

```
#include<stdio.h>
#include<math.h>

double find_Diameter(double radius);
double find_Circumference(double radius);
double find_Area(double radius);

int main()
{
    float radius, area, circumference, diameter;

    printf("\n Please Enter the radius of a circle : ");
    scanf("%f",&radius);

    diameter = find_Diameter(radius);
    circumference = find_Circumference(radius);
    area = find_Area(radius);

    printf("\n Diameter Of a Circle = %.2f\n", diameter);
    printf(" Circumference Of a Circle = %.2f\n", circumference);
    printf(" Area Of a Circle = %.2f\n", area);

    return 0;
}

double find_Diameter(double radius)
{
```

```

    return 2 * radius;
}

double find_Circumference(double radius)
{
    return 2* M_PI * radius;
}

double find_Area(double radius)
{
    return M_PI * radius * radius;
}

```

The screenshot displays the Codeboard IDE interface. The top navigation bar includes a search icon, a home icon, a project icon, and a run icon. The main workspace shows a file named 'main.c' with the following C code:

```

1 double find_Diameter(double radius);
2 double find_Circumference(double radius);
3 double find_Area(double radius);
4
5 int main()
6 {
7     float radius, area, circumference, diameter;
8
9     printf("\n Please Enter the radius of a circle : ");
10    scanf("%f",&radius);
11
12    diameter = find_Diameter(radius);
13    circumference = find_Circumference(radius);
14    area = find_Area(radius);
15
16    printf("\n Diameter Of a Circle = %.2f\n", diameter);
17    printf("\n Circumference Of a Circle = %.2f\n", circumference);
18    printf("\n Area Of a Circle = %.2f\n", area);
19
20    return 0;
21 }
22
23 double find_Diameter(double radius)
24 {
25     return 2 * radius;
26 }
27
28 double find_Circumference(double radius)
29 {
30     return 2 * M_PI * radius;
31 }
32
33 double find_Area(double radius)
34 {
35     return M_PI * radius * radius;
36 }

```

The output window at the bottom shows the results of the program execution for a radius of 9:

```

Please Enter the radius of a circle : 9
Diameter Of a Circle = 18.00
Circumference Of a Circle = 56.55
Area Of a Circle = 254.47

```

The footer of the IDE shows the user 'ashen' and their role as 'Project owner'.

Write a C program to find maximum and minimum between two numbers using functions.

```
#include <stdio.h>

int max(int num1, int num2);
int min(int num1, int num2);

int main()
{
    int num1, num2, maximum, minimum;
    printf("Enter frist numbers: ");
    scanf(" %d", &num1);

    printf("Enter secound numbers: ");
    scanf(" %d", &num2);

    maximum = max(num1, num2);
    minimum = min(num1, num2);

    printf("\nMaximum = %d\n", maximum);
    printf("Minimum = %d", minimum);

    return 0;
}

int max(int num1, int num2)
{
    return (num1 > num2 ) ? num1 : num2;
}

int min(int num1, int num2)
{
    return (num1 > num2 ) ? num2 : num1;
}
```

Codeboard - the IDE for the X

codeboard.io/projects/252218

code()

Project

Edit

View

Actions

Compile

Run

assignment c

main.c

Root

main.c

```
3 int max(int num1, int num2);
4 int min(int num1, int num2);
5
6 int main()
7 {
8     int num1, num2, maximum, minimum;
9
10    printf("Enter frist numbers: ");
11    scanf("%d",&num1);
12
13    printf("Enter secound numbers: ");
14    scanf("%d",&num2);
15
16    maximum = max(num1, num2);
17    minimum = min(num1, num2);
18
19    printf("\nMaximum = %d\n", maximum);
20    printf("\nMinimum = %d", minimum);
21
22    return 0;
23 }
24 int max(int num1, int num2)
25 {
26     return (num1 > num2 ) ? num1 : num2;
27 }
28
29 int min(int num1, int num2)
30 {
```

Enter frist numbers: 10

Enter secound numbers: 20

Maximum = 20

Minimum = 10

Activate Windows

Go to Settings to activate Windows.

Send

User: ashen

Role: Project owner

codeboard.io

Write a C program to check whether a number is even or odd using functions.

```
#include <stdio.h>

char* odd_even(int i);

int main()
{
    int number;

    printf("Enter an integer : ");
    scanf("%d", &number);

    printf("Result : %s", odd_even(number));
    return 0;
}

char* odd_even(int number)
{
    if (number%2 == 0){
        return "YOUR NUMBER IS EVEN NUMBER";
    }else{
        return "YOUR NUMBER IS ODD NUMBER";
    }
}
```

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code() Project Edit View Actions Compile Run

c programmer odd and even n

Rootmain.c

```
1 #include <stdio.h>
2
3 char* odd_even(int i);
4
5 int main()
6 {
7     int number;
8
9     printf("Enter an integer : ");
10    scanf("%d", &number);
11
12    printf("Result : %s",odd_even(number));
13    return 0;
14 }
15
16 char* odd_even(int number)
17 {
18     if (number%2 == 0){
19         return "YOUR NUMBER IS EVEN NUMBER";
20     }else{
21         return "YOUR NUMBER IS ODD NUMBER";
22     }
23 }
```

Enter an integer : 25
Result : YOUR NUMBER IS ODD NUMBER

Activate Windows
Go to Settings to activate Windows.

25 Send

User: ashen Role: Project owner

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Write a C program to check whether a number is prime, Armstrong or perfect number using functions.

```
#include <stdio.h>
#include <math.h>

int isPrime(int num);
int isArmstrong(int num);
int isPerfect(int num);

int main()
{
    int num;

    printf("Enter any number: ");
    scanf("%d", &num);

    if(isPrime(num))
    {
        printf("%d is Prime number.\n", num);
    }
    else
    {
        printf("%d is not Prime number.\n", num);
    }

    if(isArmstrong(num))
    {
        printf("%d is Armstrong number.\n", num);
    }
}
```

```

else
{
    printf("%d is not Armstrong number.\n", num);
}

if(isPerfect(num))
{
    printf("%d is Perfect number.\n", num);
}
else
{
    printf("%d is not Perfect number.\n", num);
}

return 0;
}

```

```

int isPrime(int num)
{
    int i;

    for(i=2; i<=num/2; i++)
    {

        if(num%i == 0)
        {
            return 0;
        }
    }

    return 1;
}

```

```
int isArmstrong(int num)
{
    int lastDigit, sum, originalNum, digits;
    sum = 0;

    originalNum = num;

    digits = (int) log10(num) + 1;

    while(num > 0)
    {
        lastDigit = num % 10;
        sum = sum + round(pow(lastDigit, digits));

        num = num / 10;
    }

    return (originalNum == sum);
}
```

```
int isPerfect(int num)
{
    int i, sum, n;
    sum = 0;
    n = num;

    for(i=1; i<n; i++)
    {
        if(n%i == 0)
        {
            sum += i;
        }
    }
}
```

```

    }

}

return (num == sum);
}

```

The screenshot shows the Codeboard IDE interface. The browser address bar displays `codeboard.io/projects/252218`. The IDE has a menu bar with `Project`, `Edit`, `View`, and `Actions`, along with `Compile` and `Run` buttons. On the left, a file explorer shows a project named `asingment c` with a file `main.c`. The main editor displays the following C code:

```

5 int isArmstrong(int num);
6 int isPerfect(int num);
7
8
9 int main()
10 {
11     int num;
12
13     printf("Enter any number: ");
14     scanf("%d", &num);
15
16     if(isPrime(num))
17     {
18         printf("%d is Prime number.\n", num);
19     }
20     else
21     {
22         printf("%d is not Prime number.\n", num);
23     }
24
25     if(isArmstrong(num))
26     {
27         printf("%d is Armstrong number.\n", num);
28     }
29     else
30     {
31         printf("%d is not Armstrong number.\n", num);
32     }
33 }

```

Below the code editor, the output window shows the results of running the program with the input `2`:

```

Enter any number: 2
2 is Prime number.
2 is Armstrong number.
2 is not Perfect number.

```

An `Activate Windows` watermark is visible in the bottom right corner of the IDE window. The status bar at the bottom indicates the user is `ashen` with the role of `Project owner`.

Write a C program to find all prime numbers between given interval using functions.

```
#include <stdio.h>

int isPrime(int num);
void printPrimes(int lowerLimit, int upperLimit);

int main()
{
    int lowerLimit, upperLimit;

    printf("Enter the lower and upper limit to list primes: ");
    scanf("%d%d", &lowerLimit, &upperLimit);

    printPrimes(lowerLimit, upperLimit);

    return 0;
}

void printPrimes(int lowerLimit, int upperLimit)
{
    printf("All prime number between %d to %d are: ", lowerLimit,
upperLimit);

    while(lowerLimit <= upperLimit)
    {
        if(isPrime(lowerLimit))
        {
            printf("%d, ", lowerLimit);
        }

        lowerLimit++;
    }
}
```

```

    }

}

int isPrime(int num)
{
    int i;

    for(i=2; i<=num/2; i++)
    {

        if(num % i == 0)
        {
            return 0;
        }
    }

    return 1;
}

```

The screenshot displays the Codeboard IDE interface. The top bar shows the project name 'assignment c' and the file 'main.c'. The code editor contains the following C program:

```

1  int isPrime(int num);
2  void printPrimes(int lowerLimit, int upperLimit);
3
4  int main()
5  {
6      int lowerLimit, upperLimit;
7
8      printf("Enter the lower and upper limit to list primes: ");
9      scanf("%d%d", &lowerLimit, &upperLimit);
10     printPrimes(lowerLimit, upperLimit);
11     return 0;
12 }
13
14 void printPrimes(int lowerLimit, int upperLimit)
15 {
16     printf("All prime number between %d to %d are: ", lowerLimit, upperLimit);
17     while(lowerLimit <= upperLimit)
18     {
19         if(isPrime(lowerLimit))
20         {
21             printf("%d, ", lowerLimit);
22         }
23     }
24 }

```

The output window at the bottom shows the execution results:

```

Enter the lower and upper limit to list primes: 10 20
All prime number between 10 to 20 are: 11, 13, 17, 19,

```

The status bar at the bottom indicates the user is 'ashen' and the role is 'Project owner'. The Codeboard logo is visible in the bottom right corner.

Write a C program to print all strong numbers between given interval using functions.

```
#include <stdio.h>

long long fact(int num);

void printStrongNumbers(int start, int end);


int main()
{
    int start, end;

    printf("Enter the lower limit to find strong number: ");
    scanf("%d", &start);
    printf("Enter the upper limit to find strong number: ");
    scanf("%d", &end);

    printf("All strong numbers between %d to %d are: \n", start, end);
    printStrongNumbers(start, end);

    return 0;
}

void printStrongNumbers(int start, int end)
{
    long long sum;
    int num;

    while(start != end)
    {
        sum = 0;
        num = start;
```

```

        while(num != 0)
        {
            sum += fact(num % 10);
            num /= 10;
        }

        if(start == sum)
        {
            printf("%d, ", start);
        }

        start++;
    }
}

long long fact(int num)
{
    if(num == 0)
        return 1;
    else
        return (num * fact(num-1));
}

```

The screenshot shows the CodeBoard IDE interface. The top bar indicates the project is 'assignment.c' and the file being edited is 'main.c'. The code in the editor matches the provided code. The output window at the bottom shows the following text:

```

Enter the lower limit to find strong number: 1
Enter the upper limit to find strong number: 100
All strong numbers between 1 to 100 are:
1, 2.

```

An 'Activate Windows' watermark is visible in the bottom right corner of the IDE window.

Write a C program to print all Armstrong numbers between given interval using functions.

```
#include <stdio.h>

int isArmstrong(int num);
void printArmstrong(int start, int end);

int main()
{
    int start, end;

    printf("Enter lower limit to print armstrong numbers: ");
    scanf("%d", &start);
    printf("Enter upper limit to print armstrong numbers: ");
    scanf("%d", &end);

    printf("All armstrong numbers between %d to %d are: \n", start,
end);
    printArmstrong(start, end);

    return 0;
}

int isArmstrong(int num)
{
    int temp, lastDigit, sum;

    temp = num;
    sum = 0;

    while(temp != 0)
```

```
{
    lastDigit = temp % 10;
    sum += lastDigit * lastDigit * lastDigit;
    temp /= 10;
}

if(num == sum)
    return 1;
else
    return 0;
}

void printArmstrong(int start, int end)
{
    while(start <= end)
    {
        if(isArmstrong(start))
        {
            printf("%d, ", start);
        }

        start++;
    }
}
```

Codeboard - the IDE for the X

codeboard.io/projects/252218

code() Project Edit View Actions Compile Run

Rootmain.c

```
1 #include <stdio.h>
2
3 int isArmstrong(int num);
4 void printArmstrong(int start, int end);
5
6
7
8 int main()
9 {
10     int start, end;
11
12     printf("Enter lower limit to print armstrong numbers: ");
13     scanf("%d", &start);
14     printf("Enter upper limit to print armstrong numbers: ");
15     scanf("%d", &end);
16
17     printf("All armstrong numbers between %d to %d are: \n", start, end);
18     printArmstrong(start, end);
19
20     return 0;
21 }
22
23 int isArmstrong(int num)
24 {
25     int temp, lastDigit, sum;
26
27     temp = num;
```

Enter lower limit to print armstrong numbers: 1
Enter upper limit to print armstrong numbers: 1000
All armstrong numbers between 1 to 1000 are:
1, 153, 370, 371, 407,

100%

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Send

User: ashen Role: Project ownercodeboard.io

Write a C program to print all perfect numbers between given interval using functions.

```
#include <stdio.h>

int isPerfect(int num);
void printPerfect(int start, int end);

int main()
{
    int start, end;

    printf("Enter lower limit to print perfect numbers: ");
    scanf("%d", &start);
    printf("Enter upper limit to print perfect numbers: ");
    scanf("%d", &end);

    printf("All perfect numbers between %d to %d are: \n", start,
end);
    printPerfect(start, end);

    return 0;
}

int isPerfect(int num)
{
    int i, sum;

    sum = 0;
    for(i=1; i<num; i++)
    {
        if(num % i == 0)
```

```
    {  
        sum += i;  
    }  
}
```

```
if(sum == num)  
    return 1;  
else  
    return 0;  
}
```

```
void printPerfect(int start, int end)  
{  
    while(start <= end)  
    {  
        if(isPerfect(start))  
        {  
            printf("%d ", start);  
        }  
  
        start++;  
    }  
}
```

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codeboard.io/projects/252218

code() Project Edit View Actions Compile Run

assignment c

Rootmain.c

```
7 int main()
8 {
9     int start, end;
10
11     printf("Enter lower limit to print perfect numbers: ");
12     scanf("%d", &start);
13     printf("Enter upper limit to print perfect numbers: ");
14     scanf("%d", &end);
15
16     printf("All perfect numbers between %d to %d are: \n", start, end);
17     printPerfect(start, end);
18
19     return 0;
20 }
21
22
23 int isPerfect(int num)
24 {
25     int i, sum;
26
27     sum = 0;
28     for(i=1; i<num; i++)
29     {
30         if(num % i == 0)
31         {
```

Enter lower limit to print perfect numbers: 10
Enter upper limit to print perfect numbers: 1000
All perfect numbers between 10 to 1000 are:
28 496

100%

Activate Windows
Go to Settings to activate Windows. Send

User: ashen Role: Project owner

Type here to search

ENG 01:25 15/06/2021

Write a C program to find power of any number using recursion.

```
#include <stdio.h>

int power(int n1, int n2);

int main()
{
    int base, a, result;
    printf("Enter base number: ");
    scanf("%d", &base);
    printf("Enter power number: ");
    scanf("%d", &a);
    result = power(base, a);
    printf("%d^%d = %d", base, a, result);
    return 0;
}

int power(int base, int a) {
    if (a != 0)
        return (base * power(base, a - 1));
    else
        return 1;
}
```

